

Global Solvability of the Weighted-Utility QBSDE of Liang–Xia–Yuan by Algebraic Coordinate Flattening

Technical verification and revision note

June 24, 2026

Status of the result. The argument below gives a *complete solution of the concrete QBSDE well-posedness problem* represented by equation (6.4) of Liang–Xia–Yuan, for every bounded market price of risk and without the smallness condition on $V(\Theta)$. It is, however, only *partial progress on the broader open problem* of developing a general theory for arbitrary genuinely coupled multidimensional quadratic BSDEs. It also does not, by itself, remove every smallness assumption from the authors’ full portfolio-equilibrium verification theorem.

Abstract

An earlier coordinate-flattening note proposed a useful mechanism for solving coupled quadratic backward stochastic differential equations (QBSDEs), but its main theorem omitted the predictable representation assumption and it did not verify that the mechanism applies to the QBSDE arising in Liang–Xia–Yuan’s weighted-utility portfolio model. We repair both defects.

First, we state and prove a corrected coordinate-flattening theorem under the Brownian predictable representation property. Second, using the exact coefficient formulas of Liang–Xia–Yuan, we show that their quadratic coefficients define a two-dimensional commutative, associative, unital algebra. Multiplication by the distinguished vector r has two distinct real eigenvalues throughout the nondegenerate parameter range. The corresponding character map $A \in GL_2(\mathbb{R})$ converts the algebra product into coordinatewise multiplication. The nonlinear transformation

$$X = \exp(A\bar{Y})$$

then changes the coupled QBSDE into two independent linear BSDEs. We obtain an explicit conditional-expectation representation and prove existence and uniqueness in $\mathcal{S}^\infty \times \mathcal{H}_{\text{BMO}}^2$ for arbitrary bounded terminal values and bounded κ , with no smallness assumption on $V(\Theta)$, the BMO norm, or the horizon.

Thus the specific analytic well-posedness gap associated with equation (6.4) is resolved. The broad website formulation concerning general coupled multidimensional QBSDEs remains open, and a separate integrability issue remains in extending the full equilibrium verification result without smallness.

1 Introduction and precise scope

Liang, Xia, and Yuan study time-inconsistent portfolio selection under nonlinear law-dependent preferences. Their weighted-utility example with random market coefficients leads to a two-dimensional coupled QBSDE, displayed as equation (6.4) in their paper [1]. They prove bounded/BMO well-posedness under a small-oscillation assumption

$$V(\Theta) := \sup_{\tau} \|\Theta - \mathbb{E}_{\tau}[\Theta]\|_{\infty} \ll 1, \quad \Theta = \int_0^T |\kappa_s|^2 \, ds,$$

and emphasize that the system is outside the standard diagonal and triangular theories. An automatically generated open-problem entry subsequently reformulated the remark as a request for a general well-posedness theory for genuinely coupled multidimensional QBSDEs [2].

The earlier solution attempt contained a valid central idea: if a quadratic generator is the pullback of a zero generator under a deterministic diffeomorphism, then the BSDE is solved by transforming a martingale. Three points required correction.

- (i) A filtered space that merely *supports* a Brownian motion need not have the predictable representation property. The proposed theorem was therefore false under its stated assumptions.
- (ii) A lower-order perturbation is admissible only when its *transformed* driver is well posed and remains inside the coordinate domain. Ordinary Lipschitz behavior in the original variables is not enough by itself.
- (iii) Most importantly, the attempt did not substitute the actual matrices C^1, C^2 , vectors c_i , and constants b_i of equation (6.4), and hence did not establish any result about the authors' system.

We correct the first two points in Sections 2–3. Sections 4–5 carry out the missing coefficient analysis and yield a global solution of the concrete QBSDE. Section 6 explains exactly what this does and does not settle.

2 Setting and the corrected coordinate theorem

Let $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ be a filtered probability space on which W is a d -dimensional Brownian motion. Throughout this note, unless explicitly stated otherwise, $\mathbb{F}$ is the $\mathbb{P}$ -augmentation of the natural filtration of W . In particular, W has the predictable representation property.

For a progressively measurable process Z with values in $\mathbb{R}^{m \times d}$, set

$$\|Z\|_{\text{BMO}}^2 := \sup_{\tau \leq T} \left\| \mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \right\|_\infty,$$

where the supremum is over all stopping times. We write $Z \in \mathcal{H}_{\text{BMO}}^2$ if this quantity is finite, and $Y \in \mathcal{S}^\infty$ if Y is essentially bounded on $[0, T] \times \Omega$.

Let $C \subset \mathbb{R}^m$ be open and convex and let $\Phi : \mathbb{R}^m \rightarrow C$ be a C^2 diffeomorphism with inverse Ψ . For $y \in \mathbb{R}^m$ and $z \in \mathbb{R}^{m \times d}$, define

$$Q_\Phi(y, z) := \frac{1}{2} D\Phi(y)^{-1} D^2\Phi(y)[z, z], \quad (1)$$

where

$$(D^2\Phi(y)[z, z])^\ell = \sum_{a=1}^d \sum_{i,j=1}^m \partial_{ij} \Phi^\ell(y) z^{i,a} z^{j,a}.$$

Theorem 2.1 (Corrected pure coordinate-flattening theorem). *For every $\xi \in L^\infty(\mathcal{F}_T; \mathbb{R}^m)$, the BSDE*

$$Y_t = \xi + \int_t^T Q_\Phi(Y_s, Z_s) ds - \int_t^T Z_s dW_s \quad (2)$$

has a unique solution $(Y, Z) \in \mathcal{S}^\infty \times \mathcal{H}_{\text{BMO}}^2$.

More precisely, if

$$M_t := \mathbb{E}_t[\Phi(\xi)] = M_0 + \int_0^t \widehat{Z}_s dW_s, \quad (3)$$

then

$$Y_t = \Psi(M_t), \quad Z_t = D\Psi(M_t) \widehat{Z}_t. \quad (4)$$

Proof. Let K be the closed essential range of ξ . It is compact. Since $\Phi(K)$ is a compact subset of the open convex set C , its convex hull

$$K_0 := \text{co}(\Phi(K))$$

is compact and satisfies $K_0 \Subset C$. Conditional expectation preserves closed convex sets, hence $M_t \in K_0$ for every fixed t . Taking the continuous martingale version in (3), first on rational times and then by continuity, gives $M_t \in K_0$ simultaneously for all t , almost surely.

The predictable representation in (3) follows from the standing filtration assumption. Moreover, for every stopping time τ ,

$$\mathbb{E}_\tau \int_\tau^T |\widehat{Z}_s|^2 ds = \mathbb{E}_\tau |M_T - M_\tau|^2 \leq 4 \|M\|_\infty^2.$$

Thus $\widehat{Z} \in \mathcal{H}_{\text{BMO}}^2$.

Define (Y, Z) by (4). Since M stays in $K_0 \Subset C$, the functions Ψ , $D\Psi$, and $D^2\Psi$ are bounded on the relevant range. Therefore $Y \in \mathcal{S}^\infty$ and $Z \in \mathcal{H}_{\text{BMO}}^2$. Differentiating $\Phi \circ \Psi = \text{Id}$ twice yields, for $x = \Phi(y)$ and $h = D\Phi(y)z$,

$$D^2\Psi(x)[h, h] = -D\Psi(x)D^2\Phi(y)[z, z]. \quad (5)$$

Applying Itô's formula to $Y = \Psi(M)$ and using (5) gives

$$dY_t = -Q_\Phi(Y_t, Z_t) dt + Z_t dW_t,$$

which is equivalent to (2).

For uniqueness, suppose $(\widetilde{Y}, \widetilde{Z}) \in \mathcal{S}^\infty \times \mathcal{H}_{\text{BMO}}^2$ solves (2). Then

$$\widetilde{M} := \Phi(\widetilde{Y}), \quad \widetilde{\widehat{Z}} := D\Phi(\widetilde{Y})\widetilde{Z}$$

are bounded and BMO, respectively. Itô's formula gives

$$d\widetilde{M}_t = \widetilde{\widehat{Z}}_t dW_t, \quad \widetilde{M}_T = \Phi(\xi).$$

Hence $\widetilde{M}_t = \mathbb{E}_t[\Phi(\xi)] = M_t$. The formulas for $\widetilde{Y}$ and $\widetilde{Z}$ then coincide with (4). $\square$

Remark 2.2 (Why the filtration correction is essential). *Merely assuming that the space supports W is insufficient. Let the filtration be generated by two independent Brownian motions (W, B) , take $\Phi = \text{Id}$, and set $\xi = \tanh(B_T)$. A solution of the zero-generator BSDE driven only by W would imply*

$$\xi = Y_0 + \int_0^T Z_s dW_s.$$

Conditioning on the entire path of B makes the stochastic integral have conditional mean zero, while ξ is a nonconstant B -measurable random variable. This is impossible. The filtration in Liang–Xia–Yuan is Brownian-generated, so the corrected assumption is satisfied in the application.

The following pullback identity is the appropriate rigorous replacement for an overly broad “Lipschitz lower-order perturbation” statement.

Proposition 2.3 (Pullback of a transformed BSDE). *Let $\Phi : D \rightarrow C$ be a C^2 diffeomorphism between open subsets of $\mathbb{R}^m$, with inverse Ψ . Suppose that*

$$X_t = \eta + \int_t^T g(s, X_s, H_s) ds - \int_t^T H_s dW_s \quad (6)$$

has a solution such that X remains in a compact subset of C . Put

$$Y = \Psi(X), \quad Z = D\Psi(X)H.$$

Then (Y, Z) solves the BSDE with terminal value $\Psi(\eta)$ and generator

$$f(t, y, z) = Q_\Phi(y, z) + D\Phi(y)^{-1}g(t, \Phi(y), D\Phi(y)z). \quad (7)$$

Conversely, every solution whose range lies in a compact subset of D transforms back by $X = \Phi(Y)$ and $H = D\Phi(Y)Z$. Consequently, uniqueness in either class is equivalent to uniqueness in the other.

Proof. Apply Itô's formula to $Y = \Psi(X)$ and use (5). The compact-range assumptions ensure that all derivatives used in the transformation are bounded on the relevant paths. The converse follows by applying Itô's formula to $X = \Phi(Y)$. $\square$

3 An algebraic criterion for exponential flattening

Let $\circ : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a bilinear product. For $x \in \mathbb{R}^m$, let L_x denote left multiplication,

$$L_x y := x \circ y.$$

Write $u \odot v$ for coordinatewise multiplication.

Lemma 3.1 (Simple-spectrum splitting). *Assume that $(\mathbb{R}^m, \circ)$ is a commutative, associative, unital real algebra with unit e . Suppose that for some $r \in \mathbb{R}^m$, the operator L_r has m distinct real eigenvalues $v_1, \dots, v_m$. Then there is an $A \in GL_m(\mathbb{R})$ such that*

$$A(x \circ y) = (Ax) \odot (Ay), \quad Ae = \mathbf{1}, \quad Ar = (v_1, \dots, v_m)^\top. \quad (8)$$

One concrete construction is as follows. Let $a_k^\top$ be a left eigenvector of L_r with eigenvalue v_k , normalized by $a_k^\top e = 1$, and let the rows of A be $a_k^\top$.

Proof. Associativity and commutativity give

$$L_x L_y = L_{x \circ y} = L_y L_x.$$

Fix a normalized left eigenvector $a_k^\top$ of L_r . For every x ,

$$(a_k^\top L_x) L_r = a_k^\top L_x L_r = a_k^\top L_r L_x = v_k a_k^\top L_x.$$

Because the left eigenspace associated with the simple eigenvalue v_k is one-dimensional, there is a scalar $\chi_k(x)$ such that

$$a_k^\top L_x = \chi_k(x) a_k^\top.$$

The normalization is legitimate: if $a_k^\top e = 0$, then $a_k^\top x = a_k^\top L_x e = 0$ for every x , contradicting $a_k \neq 0$. Evaluating the preceding identity at e now gives $\chi_k(x) = a_k^\top x$. Therefore

$$a_k^\top (x \circ y) = a_k^\top L_x y = (a_k^\top x) (a_k^\top y).$$

Stacking these identities proves the first relation in (8). Left eigenvectors corresponding to distinct eigenvalues are linearly independent, so A is invertible. The remaining relations follow from the normalization and

$$a_k^\top r = a_k^\top L_r e = v_k a_k^\top e = v_k.$$

$\square$

Corollary 3.2 (Exponential flattening). *Under the assumptions of Lemma 3.1, define*

$$\Phi(y) := \exp(Ay) = (e^{(Ay)_1}, \dots, e^{(Ay)_m})^\top.$$

Then $\Phi : \mathbb{R}^m \rightarrow (0, \infty)^m$ is a C^∞ diffeomorphism and

$$Q_\Phi(y, z) = \frac{1}{2} z \circ z, \tag{9}$$

where, for a d -dimensional Brownian motion, $z \circ z := \sum_{a=1}^d z_{\cdot a} \circ z_{\cdot a}$.

Proof. Since

$$D\Phi(y) = \text{diag}(e^{Ay})A, \quad D^2\Phi(y)[z, z] = \text{diag}(e^{Ay}) \begin{pmatrix} |(Az)^1|^2 \\ \vdots \\ |(Az)^m|^2 \end{pmatrix},$$

we obtain

$$Q_\Phi(y, z) = \frac{1}{2} A^{-1} \begin{pmatrix} |(Az)^1|^2 \\ \vdots \\ |(Az)^m|^2 \end{pmatrix}.$$

Equation (8) identifies this vector with $\frac{1}{2} z \circ z$. □

4 The coefficient algebra of the weighted-utility QBSDE

We now specialize to the system in Section 6 and Appendix C.2 of Liang–Xia–Yuan [1]. Their exposition takes a one-dimensional Brownian motion for notational simplicity; we do the same. The argument extends verbatim to vector Brownian motion by replacing products with Euclidean inner products.

The preference parameters satisfy

$$-1 < \gamma \leq 0, \quad \gamma \leq \rho < \gamma + 1. \tag{10}$$

The corner $(\gamma, \rho) = (0, 0)$ is the expected-utility degeneracy and makes the matrix used in equation (6.4) singular. We therefore assume

$$(\gamma, \rho) \neq (0, 0). \tag{11}$$

Define

$$r := \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} := \begin{pmatrix} \gamma \\ 1 - \rho + \gamma \end{pmatrix}, \quad \lambda := \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} := \frac{1}{1 - \rho} \begin{pmatrix} -\gamma \\ 1 - \rho + \gamma \end{pmatrix}. \tag{12}$$

Then

$$\lambda_1 + \lambda_2 = 1, \quad \delta := 1 - \lambda^\top r = \rho - 2\gamma > 0. \tag{13}$$

Let

$$P := I - r\lambda^\top, \quad Q_i := E_i - r_i\lambda\lambda^\top, \quad C^i := P^{-\top} Q_i P^{-1}, \quad i = 1, 2, \tag{14}$$

where $E_1 = \text{diag}(1, 0)$ and $E_2 = \text{diag}(0, 1)$. The matrix determinant lemma and (13) give $\det P = \delta$.

Following Appendix C.2 of [1], set

$$c_i := 2C^i r \in \mathbb{R}^2, \quad b_i := r^\top C^i r + r_i, \quad i = 1, 2. \tag{15}$$

The QBSDE is

$$\begin{cases} d\bar{Y}_t^i = -\frac{1}{2} \left[\bar{Z}_t^\top C^i \bar{Z}_t + c_i^\top \bar{Z}_t \kappa_t + b_i |\kappa_t|^2 \right] dt + \bar{Z}_t^i dW_t, & i = 1, 2, \\ \bar{Y}_T = \xi. \end{cases} \quad (16)$$

The paper has $\xi = 0$; allowing arbitrary bounded ξ costs nothing.

Define a bilinear product on $\mathbb{R}^2$ by

$$(x \circ y)_i := x^\top C^i y, \quad i = 1, 2. \quad (17)$$

Then (15) immediately gives

$$\frac{1}{2} \left(\bar{Z}^\top C^1 \bar{Z} + c_1^\top \bar{Z} \kappa + b_1 \kappa^2 \right) = \frac{1}{2} \bar{Z} \circ \bar{Z} + \kappa(\bar{Z} \circ r) + \frac{1}{2} \kappa^2(r \circ r + r). \quad (18)$$

Thus the entire generator, including its linear and constant terms, is governed by one product and one distinguished vector r .

Lemma 4.1 (Explicit product in P -coordinates). *Let $u = P^{-1}x$, $v = P^{-1}y$, and*

$$\alpha := \frac{\lambda_1 \lambda_2}{\delta}. \quad (19)$$

Then

$$P^{-1}(x \circ y) = u \star v := u \odot v + \alpha r(u_1 - u_2)(v_1 - v_2). \quad (20)$$

The product $\star$ is commutative, associative, and has unit $\mathbf{1} = (1, 1)^\top$. Consequently, $\circ$ is commutative, associative, and has unit

$$e = P\mathbf{1} = \mathbf{1} - r = \begin{pmatrix} 1 - \gamma \\ \rho - \gamma \end{pmatrix}. \quad (21)$$

Proof. From (14),

$$(x \circ y)_i = u^\top Q_i v = u_i v_i - r_i(\lambda^\top u)(\lambda^\top v).$$

Therefore

$$x \circ y = u \odot v - rab, \quad a := \lambda^\top u, \quad b := \lambda^\top v. \quad (22)$$

The Sherman–Morrison formula gives

$$P^{-1} = I + \frac{r\lambda^\top}{\delta}.$$

Applying this to (22) yields

$$P^{-1}(x \circ y) = u \odot v + \frac{r}{\delta}(\lambda^\top(u \odot v) - ab).$$

Since $\lambda_1 + \lambda_2 = 1$,

$$\lambda^\top(u \odot v) - (\lambda^\top u)(\lambda^\top v) = \lambda_1 \lambda_2 (u_1 - u_2)(v_1 - v_2),$$

which proves (20).

It remains to prove associativity. Put $d(u) := u_1 - u_2$. Expanding both parenthesizations, the terms of order α^2 are identical and cancel, while the remaining difference is

$$\begin{aligned} (u \star v) \star w - u \star (v \star w) &= \alpha d(v) [d(u)(r \odot w) - d(w)(u \odot r)] \\ &\quad + \alpha r [d(u \odot v)d(w) - d(u)d(v \odot w)]. \end{aligned}$$

The bracket in the first line is the vector

$$r(u_1 w_2 - u_2 w_1),$$

whereas the scalar bracket in the second line equals

$$d(v)(u_2 w_1 - u_1 w_2).$$

The two lines therefore cancel exactly. Thus $\star$ is associative. Finally, $d(\mathbf{1}) = 0$ and $\mathbf{1} \odot u = u$, so $\mathbf{1}$ is its unit. Transporting the product through P gives (21). $\square$

Lemma 4.2 (Simple real spectrum). *Let $L_r x = r \circ x$. Under (10)–(11), L_r has two distinct real eigenvalues.*

Proof. The algebra isomorphism P identifies L_r with left multiplication, under $\star$, by

$$s := P^{-1}r = \frac{r}{\delta}.$$

Relative to the standard basis,

$$[L_s^*] = \begin{pmatrix} s_1 + \alpha r_1(s_1 - s_2) & -\alpha r_1(s_1 - s_2) \\ \alpha r_2(s_1 - s_2) & s_2 - \alpha r_2(s_1 - s_2) \end{pmatrix}. \quad (23)$$

Its determinant is

$$\det L_s^* = s_1 s_2 + \alpha(s_1 - s_2)(r_1 s_2 - r_2 s_1) = s_1 s_2 = \frac{r_1 r_2}{\delta^2}.$$

If $\gamma < 0$, then $r_1 = \gamma < 0$ and $r_2 = 1 - \rho + \gamma > 0$ by $\rho < \gamma + 1$. Hence $\det L_r < 0$. A real 2×2 matrix with negative determinant has two distinct real eigenvalues.

If $\gamma = 0$, nondegeneracy forces $0 < \rho < 1$. In that case $\lambda_1 = 0$, so $\alpha = 0$, and

$$s = \left(0, \frac{1 - \rho}{\rho}\right)^\top.$$

Thus (23) is diagonal with the distinct eigenvalues 0 and $(1 - \rho)/\rho$. $\square$

Corollary 4.3 (Character coordinates for the authors' coefficients). *There are $A \in GL_2(\mathbb{R})$ and distinct $v_1, v_2 \in \mathbb{R}$ such that*

$$A(x \circ y) = (Ax) \odot (Ay), \quad Ae = \mathbf{1}, \quad Ar = v := \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}. \quad (24)$$

One may compute A by taking normalized left eigenvectors of L_r as its rows.

Equivalently, if the matrix c is assembled from the columns c_i in (15), then

$$L_r = \frac{1}{2}c^\top. \quad (25)$$

Thus A can be obtained directly from the coefficient matrix printed in Appendix C.2 of [1].

Proof. Apply Lemmas 3.1, 4.1, and 4.2. For (25), note that

$$(L_r x)_i = (r \circ x)_i = x^\top C^i r = \frac{1}{2}c_i^\top x.$$

$\square$

5 Global solution and explicit representation

We can now solve (16) globally. Let A and v be as in Corollary 4.3, and set

$$a_k := \frac{1}{2}v_k(v_k + 1), \quad k = 1, 2. \quad (26)$$

Let

$$\Theta_t := \int_0^t |\kappa_s|^2 \, ds.$$

Theorem 5.1 (Global well-posedness of equation (6.4)). *Assume (10)–(11), let κ be bounded and predictable, and let $\xi \in L^\infty(\mathcal{F}_T; \mathbb{R}^2)$. Then the QBSDE (16) has a unique solution*

$$(\bar{Y}, \bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \mathcal{H}_{\text{BMO}}^2(\mathbb{R}^2).$$

No smallness condition is required on $V(\Theta)$, $\|\kappa\|_\infty$, T , or the BMO norm of the solution.

For $k = 1, 2$, define an equivalent probability measure $\mathbb{Q}^k$ by

$$\frac{d\mathbb{Q}^k}{d\mathbb{P}} = \mathcal{E}\left(\int_0^T v_k \kappa_s \, dW_s\right)_T, \quad W_t^k := W_t - \int_0^t v_k \kappa_s \, ds. \quad (27)$$

Then W^k is a $\mathbb{Q}^k$ -Brownian motion. Put $\eta := A\xi$. The solution satisfies

$$X_t^k := \exp((A\bar{Y}_t)_k) = \mathbb{E}_t^{\mathbb{Q}^k}[\exp(\eta_k + a_k(\Theta_T - \Theta_t))], \quad k = 1, 2, \quad (28)$$

and hence

$$\bar{Y}_t = A^{-1} \begin{pmatrix} \log X_t^1 \\ \log X_t^2 \end{pmatrix}. \quad (29)$$

To recover $\bar{Z}$, let

$$N_t^k := \mathbb{E}_t^{\mathbb{Q}^k}[\exp(\eta_k + a_k\Theta_T)] = N_0^k + \int_0^t \tilde{H}_s^k \, dW_s^k. \quad (30)$$

Then

$$V_t^k := \frac{\tilde{H}_t^k}{N_t^k}, \quad \bar{Z}_t = A^{-1} \begin{pmatrix} V_t^1 \\ V_t^2 \end{pmatrix}. \quad (31)$$

In the authors' equation (6.4), $\xi = 0$.

Proof. Step 1: algebraic decoupling. Write

$$U := A\bar{Y}, \quad V := A\bar{Z}.$$

Using (18) and (24), the k th transformed equation is

$$U_t^k = \eta_k + \int_t^T \left[\frac{1}{2} |V_s^k|^2 + v_k \kappa_s V_s^k + a_k |\kappa_s|^2 \right] \, ds - \int_t^T V_s^k \, dW_s. \quad (32)$$

Thus the two components are independent scalar equations.

Step 2: exponential linearization. Set

$$X^k := e^{U^k}, \quad H^k := X^k V^k.$$

Itô's formula transforms (32) into the linear BSDE

$$dX_t^k = - \left[v_k \kappa_t H_t^k + a_k |\kappa_t|^2 X_t^k \right] \, dt + H_t^k \, dW_t, \quad X_T^k = e^{\eta_k}. \quad (33)$$

Because κ is bounded, Novikov's condition holds, so the stochastic exponential in (27) is a uniformly integrable martingale and $\mathbb{Q}^k$ is equivalent to $\mathbb{P}$. The predictable representation property is preserved by this Girsanov change of measure, with W^k as the representing Brownian motion. Under $\mathbb{Q}^k$, equation (33) becomes

$$dX_t^k = -a_k |\kappa_t|^2 X_t^k dt + H_t^k dW_t^k. \quad (34)$$

To construct the solution, define directly

$$N_t^k := \mathbb{E}_t^{\mathbb{Q}^k} [\exp(\eta_k + a_k \Theta_T)].$$

This is a bounded, strictly positive $\mathbb{Q}^k$ -martingale. By the predictable representation property there is a unique locally square-integrable $\tilde{H}^k$ such that (30) holds. Set

$$X_t^k = e^{-a_k \Theta_t} N_t^k, \quad H_t^k = e^{-a_k \Theta_t} \tilde{H}_t^k.$$

Itô's formula verifies (34), and substituting $dW_t^k = dW_t - v_k \kappa_t dt$ gives (33). Pulling the factor $e^{-a_k \Theta_t}$ through the conditional expectation proves (28). Formula (31) follows because $V^k = H^k / X^k = \tilde{H}^k / N^k$.

Step 3: boundedness and positivity. Let $K := T \|\kappa\|_\infty^2$. Since η_k is bounded and $0 \leq \Theta_T - \Theta_t \leq K$, (28) yields deterministic constants $0 < m_k \leq M_k < \infty$ such that

$$m_k \leq X_t^k \leq M_k \quad \text{for all } t, \quad (35)$$

almost surely. For example, one may take

$$m_k = \exp(-\|\eta_k\|_\infty - |a_k| K), \quad M_k = \exp(\|\eta_k\|_\infty + |a_k| K).$$

Hence $U^k = \log X^k$ and $\bar{Y} = A^{-1}U$ are bounded.

Step 4: the BMO estimate. Fix k and a stopping time τ . Let $\sigma_n := \inf\{t \geq \tau : \int_\tau^t |H_s^k|^2 ds \geq n\} \wedge T$. These stopping times localize the stochastic integral in (33). Applying Itô's formula to $(X^k)^2$ on $[\tau, T \wedge \sigma_n]$, taking $\mathbb{E}_\tau$, and using (35), we obtain

$$\begin{aligned} \mathbb{E}_\tau \int_\tau^{T \wedge \sigma_n} |H_s^k|^2 ds &\leq M_k^2 + 2\mathbb{E}_\tau \int_\tau^{T \wedge \sigma_n} X_s^k v_k \kappa_s H_s^k ds + 2|a_k| \mathbb{E}_\tau \int_\tau^{T \wedge \sigma_n} |\kappa_s|^2 (X_s^k)^2 ds \\ &\leq M_k^2 + \frac{1}{2} \mathbb{E}_\tau \int_\tau^{T \wedge \sigma_n} |H_s^k|^2 ds + 2v_k^2 M_k^2 K + 2|a_k| M_k^2 K. \end{aligned}$$

Conditional Fatou's lemma as $n \rightarrow \infty$ gives

$$\mathbb{E}_\tau \int_\tau^T |H_s^k|^2 ds \leq 2M_k^2 (1 + 2v_k^2 K + 2|a_k| K).$$

The bound is deterministic and uniform in τ , so $H^k \in \mathcal{H}_{\text{BMO}}^2$. Since $X^k \geq m_k$, also $V^k = H^k / X^k \in \mathcal{H}_{\text{BMO}}^2$. Therefore $\bar{Z} = A^{-1}V \in \mathcal{H}_{\text{BMO}}^2$.

Step 5: uniqueness. Let $(\tilde{U}^k, \tilde{V}^k)$ be any bounded/BMO solution of (32). Define

$$\tilde{X}^k = e^{\tilde{U}^k}, \quad \tilde{H}^k = \tilde{X}^k \tilde{V}^k.$$

Then $\tilde{X}^k$ is bounded above and away from zero, and

$$e^{a_k \Theta_t} \tilde{X}_t^k$$

is a bounded local martingale under $\mathbb{Q}^k$. It is therefore a true martingale and must equal the conditional expectation in (30). Hence $\tilde{X}^k = X^k$. Uniqueness of the stochastic-integral representation gives $\tilde{H}^k = H^k$, and therefore $\tilde{U}^k = U^k$ and $\tilde{V}^k = V^k$. Transforming by A^{-1} proves uniqueness of $(\bar{Y}, \bar{Z})$. $\square$

Remark 5.2 (The coordinate map). *The transformation used in Theorem 5.1 is exactly the exponential flattening map*

$$\Phi(y) = \exp(Ay) : (\mathbb{R}^2, \circ) \longrightarrow (0, \infty)^2.$$

The quadratic part of equation (6.4) is Q_Φ , while its lower-order terms become the linear transformed driver

$$g_k(t, x, h) = v_k \kappa_t h + a_k |\kappa_t|^2 x.$$

The positive quadrant is invariant because of the explicit representation (28). This is the domain verification that was missing from the earlier perturbation argument.

Remark 5.3 (Vector Brownian motion). *If W and κ are d -dimensional and $\bar{Z}$ is $2 \times d$, define*

$$z \circ z := \sum_{a=1}^d z_{\cdot a} \circ z_{\cdot a}, \quad \kappa(z \circ r) := \sum_{a=1}^d \kappa^a(z_{\cdot a} \circ r).$$

The proof is unchanged. The scalar transformed equations contain $\frac{1}{2} |V^k|^2 + v_k \langle \kappa, V^k \rangle + a_k |\kappa|^2$, and the Girsanov density is $\mathcal{E}(\int v_k \kappa \cdot dW)$.

6 What has been solved, and what remains

6.1 The concrete QBSDE problem is solved

Lemma 6.3 of Liang–Xia–Yuan proves well-posedness of equation (6.4) only when $V(\Theta)$ is sufficiently small, and only in a sufficiently small BMO ball. Theorem 5.1 removes both restrictions. It supplies a unique bounded/BMO solution for every bounded κ and gives an explicit representation. In that precise sense, this note gives a complete solution of the analytic well-posedness problem for the QBSDE actually appearing in the paper.

The solution also determines the formal candidate satisfying the authors’ first-order equilibrium condition: once $\bar{Z}$ is known, their formula (6.5) defines $\bar{\pi}$, and the volatility-scaled strategy is BMO because κ is bounded and $\bar{Z} \in \mathcal{H}_{\text{BMO}}^2$. This statement is deliberately weaker than the full equilibrium verification discussed below.

6.2 The broad multidimensional problem is not solved

The generated website problem asks for general existence and uniqueness results for genuinely coupled multidimensional QBSDEs beyond diagonal and triangular forms. Our theorem identifies a nontrivial but special class: the constant quadratic coefficients must form a split semisimple commutative algebra, and the lower-order terms must be compatible with the same distinguished element r . General coupled quadratic systems need not be associative, unital, flat, or split over $\mathbb{R}$. Thus the website problem receives substantive partial progress, not a general solution.

6.3 The full equilibrium verification still has an integrability gap

The proof of Proposition 6.1 in [1] uses the small BMO norm furnished by Lemma 6.3 to obtain strong exponential-moment estimates for the candidate wealth process and its reciprocal. Theorem 5.1 proves that $\bar{Z}$ has a finite BMO norm, but it does not force that norm to be small. Standard BMO theory gives a reverse-Hölder exponent strictly larger than one, but the exponent may deteriorate as the BMO norm grows [3, Chapters 2–3]. Consequently, the high-moment estimates used in the authors’ verification argument do not follow automatically from our global QBSDE theorem.

Accordingly, the present result removes the smallness assumption from QBSDE solvability and from construction of the first-order candidate, but a separate argument is needed to remove smallness from the complete Type-II equilibrium verification.

6.4 Two minor points in the source proof

Our calculations also clarify two harmless but genuine issues in Appendix C.3 of [1].

- (a) Proposition C.1 is stated under the assumption that the matrix c has two real eigenvalues, “possibly the same,” but its proof immediately chooses two linearly independent eigenvectors. The correct hypothesis is that c is diagonalizable over $\mathbb{R}$, or, more simply for the application, that it has two distinct real eigenvalues.
- (b) In the boundary case $\gamma = 0$, the proof of Lemma 6.3 cannot conclude $\det c < 0$: in fact $\det c = 0$. Nevertheless, when $0 < \rho < 1$, one directly has

$$c = \text{diag} \left(0, \frac{2(1-\rho)}{\rho} \right),$$

so the required real diagonalization remains valid. Lemma 4.2 treats this case separately.

7 Conclusion

The coordinate-flattening idea survives rigorous scrutiny after two essential repairs: one must assume Brownian predictable representation, and one must verify that the exact coefficient system is flat and that the transformed lower-order dynamics preserve the coordinate domain. For the weighted-utility QBSDE of Liang–Xia–Yuan, these verifications can be completed exactly.

The decisive structural fact is algebraic. The matrices C^1, C^2 define a commutative, associative, unital product, and multiplication by r has simple real spectrum. Character coordinates turn the product into coordinatewise multiplication, after which the componentwise exponential transform produces two scalar linear BSDEs. This gives global existence, uniqueness, BMO control, and an explicit conditional-expectation formula without any smallness assumption.

The result is therefore a solution of the concrete QBSDE well-posedness question in equation (6.4), partial progress on the broad theory of coupled multidimensional QBSDEs, and partial rather than complete progress on removing smallness from the paper’s full portfolio-equilibrium theorem.

A A direct consistency check against the coefficient matrix c

For readers wishing to compare with Appendix C.2 of [1], recall that c is the matrix with columns c_1, c_2 . From (25),

$$c^\top = 2L_r.$$

For $\gamma < 0$, Lemma 4.2 gives

$$\det c = 4 \det L_r = \frac{4r_1 r_2}{\delta^2} = \frac{4\gamma(1-\rho+\gamma)}{(\rho-2\gamma)^2} = -\frac{4\gamma(\rho-\gamma-1)}{(\rho-2\gamma)^2} < 0.$$

This agrees with the determinant obtained by simplifying the explicit matrix in equation (C.9), but avoids expanding its individual entries.

The identity $b_i = r^\top C^i r + r_i$ and the relation $c_i = 2C^i r$ imply

$$A \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = A(r \circ r + r) = (Ar) \odot (Ar) + Ar = \begin{pmatrix} v_1^2 + v_1 \\ v_2^2 + v_2 \end{pmatrix},$$

which is precisely the constant coefficient appearing after transformation. Likewise,

$$A \begin{pmatrix} c_1^\top z \\ c_2^\top z \end{pmatrix} = 2A(z \circ r) = 2(Az) \odot v,$$

which verifies the linear coefficient in (32).

References

- [1] Z. Liang, J. Xia, and F. Yuan. Dynamic portfolio selection for nonlinear law-dependent preferences. *Mathematics of Operations Research*, 2025. DOI: [10.1287/moor.2023.0345](https://doi.org/10.1287/moor.2023.0345); arXiv:2311.06745v2.
- [2] Open Problems in Operations Research. “Well-posedness theory for coupled multidimensional quadratic BSDEs in portfolio equilibria.” https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4411801764_p0&n=18&page=2. The website marks the generated formulation as unverified.
- [3] N. Kazamaki. *Continuous Exponential Martingales and BMO*. Lecture Notes in Mathematics 1579, Springer, 1994.